LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 4: Procedures with Inputs

Module 2 | 45 minutes | Procedures and parameters | Grades 6–8 · ages 11–14

Objective: Define custom blocks (procedures) that take an input, and call them with different arguments to reuse code.

Success criteria

- I can define a procedure with a parameter.
- I can call it with different arguments.
- I can explain parameter vs argument.

Key vocabulary

- **procedure / function**: A named, reusable block of code.
- **parameter**: The input name in the definition.
- **argument**: The value passed when calling.
- **call**: Using (running) a procedure.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Function machine (5 min): a learner is “double (n)”; the class passes arguments and gets outputs.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	The polygon procedure: learners follow the steps.
14 min	**Perform**	Code Lab: polygon procedure, Match the word to the example in polygon (sides) ... polygon (5)
9 min	**Probe**	Code Lab: polygon flower, Functions in Python, activity
4 min	**Wrap-up**	Exit ticket: In polygon (8), what is the argument?

Unplugged warm-up

Function machine (5 min): a learner is “double (n)”; the class passes arguments and gets outputs.

Teaching the idea

Procedures (My Blocks / Python functions) encapsulate reusable behaviour. Parameters are placeholders in the definition; arguments are values at the call. Parameters are local to the definition.

Analogy: A coffee machine button “make coffee (size)”: the machine’s steps are fixed, you choose the size.

Live demo: Write the 3 polygons long-hand, then refactor into polygon (sides).

Questions to ask

Ask	Listen for
Where is the parameter?	In the define hat.
What is the argument in polygon (5)?	5.
Why test the procedure alone?	Find bugs before combining.

Differentiation

Support
Provide the define hat with sides; learners fill in the body.

Stretch
Make star (size) and use it in a loop with growing sizes.

Common misconceptions

Misconception	How to fix it
Defining runs the code	Only calling runs it.
Parameters are global variables	They exist only inside the define script.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
In polygon (8), what is the argument?

Homework
Write three procedures you could use in daily life, each with one parameter (e.g. walk (minutes)).

Answer key — Kitabash activities

1. Code Lab: polygon procedure (perform · code lab)

Goals checked: Defines and calls a procedure · Uses the parameter · Draws 3 + 4 + 5 sides

One solution:

Pen



2. Match the word to the example in polygon (sides) … polygon (5) (perform · matching)

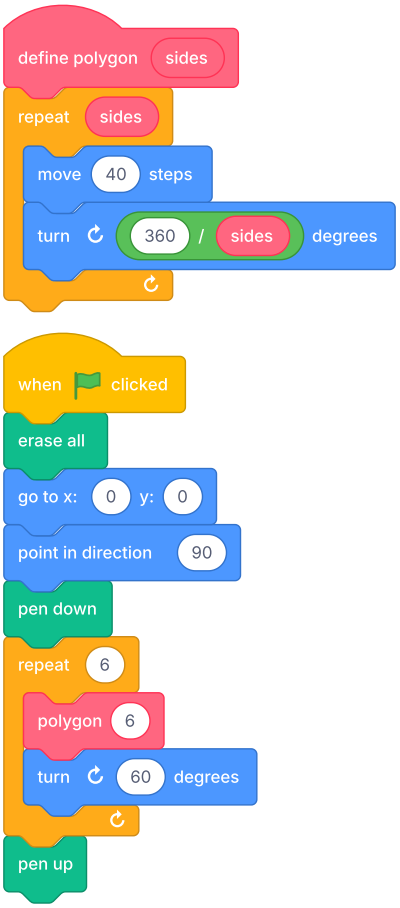
procedure → polygon · parameter → sides · argument → 5 · definition → the define polygon script

3. Code Lab: polygon flower (probe · code lab)

Goals checked: Calls the procedure · Repeats it · 36 lines drawn

One solution:

Pen



4. Functions in Python (probe · code output)

5. Explain your thinking (probe · free response)

Look for: parameter, argument, call. **Example:** In my block polygon (sides), sides is the parameter: it is the name the procedure uses inside. When I call polygon (5), the 5 is the argument: it is the actual value I pass in, so sides becomes 5 for that call.

Answer key — worksheet

1. **Key words:** procedure / function = A named, reusable block of code. · parameter = The input name in the definition. · argument = The value passed when calling. · call = Using (running) a procedure.
2. **Plan your code:** Goals: Defines and calls a procedure; Uses the parameter; Draws 3 + 4 + 5 sides.
3. **Match the word to the example in polygon (sides) … polygon (5):** procedure → polygon · parameter → sides · argument → 5 · definition → the define polygon script

4. **Plan your code:** Goals: Calls the procedure; Repeats it; 36 lines drawn.
5. **Functions in Python:** A. Hi Sam then Hi Lee

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